How far can you get with data and stats?

PSYC 11: Laboratory in Psychological Science

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What is the point of this lab exercise?

Lab Goals

- We're trying to gain insights into what makes an effective "pitch"
- Relevant to the Introduction section of scientific articles
- Also relevant to presenting/describing your science

What tools do we have to achieve our goal?

Available Tools

- Ratings data
- Analytic tools (stats!)
- Visualization tools (figures!)
- **Our intuitions**

Our dataset

Pitch ratings (W22)

File Edit View Insert Format Data Tools Extensions Help

	A	B	C	D	E	F
1	Timestamp	Which group's pitch are y	How CLEAR was the pitc	How INTERESTING was	How EFFICIENT was the	How effective was the chosen FORMAT of the pitch?
2	4/6/2022 12:59:35	B	10	10	10	10
3	4/6/2022 12:59:41	B	10	8	10	9
4	4/6/2022 12:59:51	B	8	10	7	8
5	4/6/2022 13:00:00	B	6	7	3	3
6	4/6/2022 13:00:24	B	10	10	10	10
7	4/6/2022 13:00:25	B	9	10	10	10
8	4/6/2022 13:00:25	B	8	8	8	9
9	4/6/2022 13:00:35	B	9	9	7	9
10	4/6/2022 13:00:36	B	10	10	9	10
11	4/6/2022 13:00:38	B	7	8	8	9
12	4/6/2022 13:00:54	B	10	9	7	10
13	4/6/2022 13:00:56	B	9	8	9	9
14	4/6/2022 13:00:57	B	10	10	10	10
15	4/6/2022 13:01:02	B	8	9	8	8
16	4/6/2022 13:01:03	B	9	9	8	9
17	4/6/2022 13:01:04	B	9	8	8	10
18	4/6/2022 13:01:42	B	10	10	8	9
19	4/6/2022 13:01:48	B	8	10	8	8
20	4/6/2022 13:01:50	B	8	6	7	8
21	4/6/2022 13:01:51	B	9	10	9	10
22	4/6/2022 13:02:02	B	8	6	6	6
23	4/6/2022 13:02:06	B	8	9	9	9
24	4/6/2022 13:02:11	B	7	10	9	8
25	4/6/2022 13:02:24	B	10	10	10	10
26	4/6/2022 13:02:26	B	8	2	10	8
27	4/6/2022 13:02:26	B	8	9	9	8
28	4/6/2022 13:02:26	B	7	8	7	9
29	4/6/2022 13:02:38	B	9	10	7	10
30	4/6/2022 13:03:02	B	10	10	8	9
31	4/6/2022 13:03:03	B	10	10	10	10
32	4/6/2022 13:15:08	C	10	8	10	9
33	4/6/2022 13:15:13	C	10	9	8	8
34	4/6/2022 13:15:15	C	8	10	7	6

Some things we can do with the dataset

What Data Can Tell Us

- Make plots to show different groups' ratings
- Run statistical tests

Some things we can't do with (only) the dataset

Limitations of Data Alone

- Actually know the "truth" about which presentation was best
- Get specific insights into how each pitch could be improved (or what the best parts were)
- Factor in potential sources of bias--- presentation order, mood, confounding variables, etc.

Trust your intuitions

Data + Intuition

- Ultimately, we can't fully discount data if we want to do good science
- But don't throw your intuitions out the window
- Use common sense to help interpret your results, understand limitations of your data/analysis, and figure out what you think is "really" going on
- Communicate your best understanding of the **truth**

Example visualizations and stats